

Microstrip Patch Antenna Design Using C



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Described here is a program in C for designing the microstrip patch antenna for a frequency range between 2.1 GHz and 5.6 GHz.

Antennae are just lengths of conductive metal that radiate radio signals into the air. Most common antennae are designed to be one-quarter or one-half the wavelength of the radio signal. Keeping in mind that antennae are generally one-quarter the wavelength of the radio signal, basic antennae for 900 MHz are typically no more than

7.62 cm (3 inches) in height.

Patch antenna

In telecommunication, there are several types of microstrip antennae, the most common of which are the microstrip patch antennae, also known simply as microstrip antennae or patch antennae. These are becoming increasingly popular in wireless applications due to their low-profile structure. Therefore they are extremely compatible with embedded antennae in handheld wireless devices such as cellular phones. The telemetry and communication antennae on missiles need to be thin and conformal, and are also often in the form of microstrip patch antennae. Another area where they have been used successfully is in satellite communication.

In its most fundamental form, a microstrip patch antenna consists of a radiating patch on one side of a dielectric substrate, which has a ground plane on the other side, as shown in Fig. 1. The patch is generally made of conducting material such as copper or gold, and can be of any shape. Common microstrip antenna shapes are square, rectangular, circular and elliptical, but any

continuous shape is possible.

Some of the advantages of microstrip antennae are their light weight and low volume.

Because of their low-profile planar configurations, they can be easily made conformal to the host surface. The fabrication cost is low; hence, they can be manufactured in large numbers.

They support both linear as well as circular polarisation and can be easily integrated with microwave integrated circuits (MICs).

Design parameters

Rectangular, single polarisation microstrip antennae are the most popular, for which there are three essential design parameters: frequency of operation, dielectric constant of the substrate and the height of the substrate.

Frequency of operation (f_o). The resonant frequency of the antenna must be selected appropriately. Some mobile communication systems operate in the frequency range of 2100 to 5600 MHz. Hence, the antenna designed must be able to operate in this frequency range.

Dielectric constant of the substrate (ϵ_r). The dielectric material selected for our design has a dielectric constant of 4.8. A substrate with a high dielectric constant has been selected, since it reduces the dimensions of the antenna.

Height of dielectric substrate (h). For the microstrip patch antenna to be

used in cellular phones, it is essential that the antenna is compact and not bulky.

The design is based on standard relationships listed in Box 1.

Software program

The software program is designed in C language.

Box 1: Designing the patch

Calculation of the actual length of patch (L)
 $L = L_{eff} - 2\Delta L$

Calculation of the width of patch (W)

$$W = \frac{c}{2f_o \sqrt{(\epsilon_r + 1)/2}}$$

Calculation of the effective dielectric constant (ϵ_{eff})

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \sqrt{\frac{1}{1 + 12 \frac{h}{W}}}$$

Calculation of the effective length (L_{eff})

$$L_{eff} = \frac{c}{2f_o \sqrt{\epsilon_{eff}}}$$

Calculation of the length extension (ΔL)

$$\Delta L = 0.412h \left(\frac{\epsilon_{eff} + 0.3}{\epsilon_{eff} - 0.258} \right) \left(\frac{W}{h} + 0.264 \right) \left(\frac{W}{h} + 0.813 \right)$$

Calculation of the ground plane dimensions (Lg & Wg)

$$L_g = L + 6h$$

$$W_g = W + 6h$$

Calculation of the feed point location (Yf & Xf)

$$Xf = L / (2\sqrt{\epsilon_{eff}})$$

$$Yf = W/2$$

where:

f_o = Frequency of operation

ϵ_r = Dielectric constant

c = Speed of Light (3×10^8 m/sec)

h = Height of dielectric constant

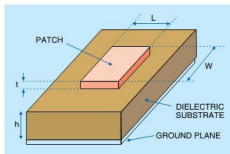


Fig. 1: Microstrip patch antenna